ECON 5140: Applied Econometrics

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Road Safety and Accident Severity in the United States: A Case Study of Texas

Road traffic accidents create serious economic and social burdens through medical costs, vehicle damage, traffic delays, lower productivity, and loss of life. Although reducing crashes is a key policy goal, it is equally important to understand why some accidents are far more severe than others. Two crashes may both appear in official records, yet the outcomes can differ substantially depending on surrounding conditions and the road environment. Studying accident severity can therefore provide useful guidance for transportation policy and emergency response.

Weather and roadway context are likely to shape crash outcomes. Rain, poor visibility, and strong wind can reduce driver control and increase roadway risk. Intersections, junctions, traffic lights, and pedestrian crossings may also influence driver behavior and collision type. This paper examines how these factors are related to accident severity in the United States, using the state of Texas as the main case study. Texas is a useful setting because it includes major cities, highway systems, rural roads, and changing weather patterns.

The study uses several econometric methods. Ordinary least squares (OLS) is estimated as a benchmark model using the four level severity scale, while ordered logit is the main specification because accident severity is naturally ordinal. A binary logit model is also used to estimate the probability of a high severity crash. Across models, rain, higher wind speed, and junction locations are positively associated with greater crash severity, while traffic signals and pedestrian crossings are associated with lower severity outcomes. Overall, the results suggest that environmental conditions and roadway design are closely related to accident severity.

The data used in this study come from the U.S. Accidents dataset, a large public source containing detailed traffic accident records from 2016 to 2023. Because the file is very large, Texas is selected as the primary case study, and a random sample of 20,000 observations is used for estimation. The dataset includes information on accident severity, location, time, weather conditions, visibility, and roadway features. Key variables include rain, visibility, temperature,

wind speed, nighttime conditions, weekends, rush hour periods, junction locations, traffic signals, and pedestrian crossings. County and month controls are included in selected specifications.

The study uses several econometric methods to examine how weather conditions and roadway context are related to accident severity. Let $Severity_i \in \{1,2,3,4\}$ denote the severity level of accident i, where higher values represent more serious incidents. As an initial benchmark, ordinary least squares is estimated as

$$Severity_i = \alpha + \beta X_i + \gamma County_i + \delta Month_i + \varepsilon_i$$

where X_i includes rain, visibility, temperature, wind speed, nighttime conditions, weekends, rush hour periods, junction locations, traffic signals, and pedestrian crossings. County and month controls are included to capture geographic and seasonal differences, and robust standard errors are reported. Since the dependent variable is ordered rather than continuous, the main specification is an ordered logit model. A latent severity index is written as

$$Severity_i^* = \alpha + \beta X_i + u_i$$

and the observed outcome takes values one through four according to estimated threshold values. This approach is appropriate because it preserves the ranking of severity categories without imposing equal spacing between them.

In addition, a binary logit model is estimated for high severity crashes, where $HighSeverity_i =$ 1if severity equals three or four, and zero otherwise. The probability model is given by

$$P(HighSeverity_i = 1 \mid X_i) = \frac{e^{\alpha+\beta X_i}}{1 + e^{\alpha+\beta X_i}}$$

Marginal effects are reported to measure changes in predicted probabilities. County clustered standard errors are used as an additional robustness check to allow correlation among accidents within the same county. All estimates are interpreted as conditional relationships rather than causal effects, since weather and roadway conditions are not randomly assigned.

The results are broadly consistent across model specifications and indicate that environmental conditions and roadway design are meaningful predictors of accident severity. Main regression estimates are reported in *Tables 1 through 3*. In the benchmark OLS model shown in *Table 1*, the regression uses 20,000 observations and explains 8.8 percent of the variation in accident severity.

Rain and wind speed are positively associated with more severe crashes, while junction locations are linked to higher severity outcomes. By contrast, traffic signals and pedestrian crossings are associated with lower severity levels. *Table 2* reports ordered logit estimates and reaches the same general conclusions. *Table 3* presents binary logit estimates for high severity crashes and confirms the same overall pattern. The consistency across models suggests that environmental conditions and roadway design are meaningful predictors of accident severity.

The binary logit estimates reinforce these findings when attention is limited to severe crashes. *Figure 1* presents marginal effects from that model. Junction locations produce one of the largest positive effects on severe crash probability, while traffic signals and pedestrian crossings reduce that probability. Rain is also associated with a higher likelihood of severe crashes. *Figure 2* further shows that the share of high severity crashes is higher during rainy conditions than during dry conditions.

Additional checks indicate that the core findings are stable. *Table 4* reports similar coefficient patterns after clustering standard errors at the county level. *Figure 3* compares Texas with the broader national sample and shows that the state generally follows national severity patterns. *Figure 4* shows that the direction of the principal coefficients remains similar across models. Taken together, the evidence suggests that weather conditions and roadway infrastructure are closely connected to accident severity and may be relevant for traffic safety planning, signal placement, and roadway design.

Robustness checks are used to test whether the baseline findings depend on model assumptions. *Table 4* reports estimates with county clustered standard errors, where the main coefficients remain close to baseline values. Rain continues to increase severity, while wind speed remains positive. Junction locations remain one of the strongest predictors, whereas traffic signals and pedestrian crossings remain negative. A stricter outcome measure based only on the highest severity category produces the same general pattern, with junction locations still positive, traffic signals negative, and rain positive but smaller in magnitude. Overall, the principal results remain stable across alternative inference methods and outcome definitions.

Several limitations should be considered when interpreting the results. The estimated coefficients describe conditional relationships rather than causal effects, since weather conditions and roadway environments are not randomly assigned. Important determinants of crash severity,

including vehicle speed, driver behavior, traffic volume, and emergency response time, are not directly observed in the dataset and may influence the estimated relationships. The sample consists of 20,000 reported Texas accidents, so unreported incidents are not captured. Severity classifications may also vary across counties or reporting agencies. While Texas provides substantial geographic and roadway variation, the estimated patterns may not fully generalize to states with different traffic density, climate, or infrastructure.

This paper examined how weather conditions and roadway context are related to accident severity in the United States, using Texas as the primary case study. Across multiple econometric specifications, rain, higher wind speed, and junction locations were consistently associated with more severe accident outcomes, while traffic signals and pedestrian crossings were associated with lower severity levels. These relationships remained stable under alternative specifications and clustered standard errors, indicating that the main findings are not driven by a single model choice. The results suggest that crash severity is influenced not only by driver behavior, but also by environmental and infrastructure conditions. Investments in safer intersections, effective traffic controls, and targeted responses during adverse weather may help reduce the number of serious crashes and their economic costs.

Appendix

Table 1 - OLS Regression Results

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                            OLS Regression Results
==============================================================================
Dep. Variable:               Severity   R-squared:                       0.088
Model:                            OLS   Adj. R-squared:                  0.086
Method:                 Least Squares   F-statistic:                     89.17
Date:                Fri, 17 Apr 2026   Prob (F-statistic):               0.00
Time:                        19:50:53   Log-Likelihood:                -11984.
No. Observations:               20000   AIC:                         2.404e+04
Df Residuals:                   19963   BIC:                         2.434e+04
Df Model:                          36
Covariance Type:                  HC1
==========================================================================================
                                     coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
Intercept                          2.2158      0.028     79.577      0.000       2.161       2.270
Junction[T.True]                   0.1482      0.018      8.192      0.000       0.113       0.184
Traffic_Signal[T.True]            -0.0625      0.007     -8.689      0.000      -0.077      -0.048
Crossing[T.True]                  -0.1245      0.008    -15.655      0.000      -0.140      -0.109
C(County_Grouped)[T.Collin]       -0.1484      0.029     -5.065      0.000      -0.206      -0.091
C(County_Grouped)[T.Dallas]        0.0337      0.015      2.236      0.025       0.004       0.063
C(County_Grouped)[T.Denton]        0.2235      0.040      5.627      0.000       0.146       0.301
C(County_Grouped)[T.El Paso]      -0.2306      0.021    -10.811      0.000      -0.272      -0.189
C(County_Grouped)[T.Fort Bend]    -0.0549      0.058     -0.948      0.343      -0.168       0.059
C(County_Grouped)[T.Harris]       -0.0610      0.014     -4.234      0.000      -0.089      -0.033
C(County_Grouped)[T.Hays]          0.2112      0.074      2.865      0.004       0.067       0.356
C(County_Grouped)[T.Montgomery]    0.2902      0.063      4.604      0.000       0.167       0.414
C(County_Grouped)[T.Other]         0.2449      0.033      7.483      0.000       0.181       0.309
C(County_Grouped)[T.Rockwall]      0.3737      0.091      4.129      0.000       0.196       0.551
C(County_Grouped)[T.Smith]        -0.2706      0.016    -17.351      0.000      -0.301      -0.240
C(County_Grouped)[T.Tarrant]       0.0979      0.021      4.709      0.000       0.057       0.139
C(County_Grouped)[T.Travis]       -0.1450      0.015     -9.981      0.000      -0.173      -0.117
C(County_Grouped)[T.Wichita]      -0.2732      0.014    -18.848      0.000      -0.302      -0.245
C(County_Grouped)[T.Williamson]   -0.0819      0.034     -2.413      0.016      -0.148      -0.015
C(Month)[T.2]                      0.0130      0.014      0.910      0.363      -0.015       0.041
C(Month)[T.3]                      0.0391      0.015      2.558      0.011       0.009       0.069
C(Month)[T.4]                      0.0332      0.016      2.052      0.040       0.001       0.065
C(Month)[T.5]                      0.0424      0.018      2.387      0.017       0.008       0.077
C(Month)[T.6]                      0.0352      0.019      1.896      0.058      -0.001       0.072
C(Month)[T.7]                      0.0053      0.018      0.291      0.771      -0.030       0.041
C(Month)[T.8]                      0.0176      0.017      1.016      0.310      -0.016       0.052
C(Month)[T.9]                      0.0193      0.016      1.171      0.241      -0.013       0.052
C(Month)[T.10]                     0.0344      0.015      2.274      0.023       0.005       0.064
C(Month)[T.11]                    -0.0147      0.013     -1.110      0.267      -0.041       0.011
C(Month)[T.12]                    -0.0115      0.013     -0.869      0.385      -0.037       0.014
Q("Visibility(mi)")            -1.855e-05      0.002     -0.012      0.991      -0.003       0.003
Rain                               0.0556      0.016      3.518      0.000       0.025       0.087
Q("Temperature(F)")                0.0004      0.000      1.368      0.171      -0.000       0.001
Q("Wind_Speed(mph)")               0.0020      0.001      2.899      0.004       0.001       0.003
Night                              0.0018      0.008      0.228      0.820      -0.014       0.018
RushHour                          -0.0035      0.007     -0.523      0.601      -0.017       0.010
Weekend                            0.0144      0.010      1.508      0.132      -0.004       0.033
==============================================================================
Omnibus:                     3810.335   Durbin-Watson:                   1.995
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             6728.875
Skew:                           1.228   Prob(JB):                         0.00
Kurtosis:                       4.428   Cond. No.                     1.56e+03
==============================================================================
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Table 2 - Ordered Logit Results

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                             OrderedModel Results
==============================================================================
Dep. Variable:                Severity   Log-Likelihood:                -12253.
Model:                    OrderedModel   AIC:                         2.453e+04
Method:             Maximum Likelihood   BIC:                         2.461e+04
Date:                 Fri, 17 Apr 2026
Time:                         19:52:30
No. Observations:                20000
Df Residuals:                    19990
Df Model:                            7
==================================================================================
                     coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
Visibility(mi)     0.0080      0.009      0.925      0.355      -0.009       0.025
Rain               0.3697      0.078      4.766      0.000       0.218       0.522
Temperature(F)     0.0025      0.001      2.222      0.026       0.000       0.005
Wind_Speed(mph)    0.0262      0.003      7.547      0.000       0.019       0.033
Night              0.0721      0.042      1.698      0.089      -0.011       0.155
RushHour           0.0222      0.037      0.607      0.544      -0.050       0.094
Weekend            0.1637      0.049      3.356      0.001       0.068       0.259
1/2               -4.4272      0.142    -31.215      0.000      -4.705      -4.149
2/3                1.8294      0.014    132.768      0.000       1.802       1.856
3/4                1.1578      0.020     57.252      0.000       1.118       1.197
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Table 3 - Binary Logit Results

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                           Logit Regression Results
==============================================================================
Dep. Variable:           HighSeverity   No. Observations:                20000
Model:                          Logit   Df Residuals:                    19978
Method:                           MLE   Df Model:                           21
Date:                Fri, 17 Apr 2026   Pseudo R-squ.:                 0.04213
Time:                        19:53:54   Log-Likelihood:                -10138.
converged:                       True   LL-Null:                       -10584.
Covariance Type:            nonrobust   LLR p-value:                 3.029e-175
==============================================================================================
                                coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Intercept                    -1.4361      0.140    -10.270      0.000      -1.710      -1.162
Junction[T.True]              0.7113      0.065     10.862      0.000       0.583       0.840
Traffic_Signal[T.True]       -0.4921      0.049     -9.961      0.000      -0.589      -0.395
Crossing[T.True]             -1.1403      0.087    -13.077      0.000      -1.311      -0.969
C(Month)[T.2]                 0.0703      0.086      0.820      0.412      -0.098       0.238
C(Month)[T.3]                 0.2721      0.088      3.081      0.002       0.099       0.445
C(Month)[T.4]                 0.2799      0.089      3.139      0.002       0.105       0.455
C(Month)[T.5]                 0.4279      0.095      4.526      0.000       0.243       0.613
C(Month)[T.6]                 0.4243      0.100      4.242      0.000       0.228       0.620
C(Month)[T.7]                 0.3284      0.102      3.212      0.001       0.128       0.529
C(Month)[T.8]                 0.3519      0.099      3.557      0.000       0.158       0.546
C(Month)[T.9]                 0.2622      0.095      2.747      0.006       0.075       0.449
C(Month)[T.10]                0.2039      0.088      2.321      0.020       0.032       0.376
C(Month)[T.11]               -0.0916      0.084     -1.089      0.276      -0.256       0.073
C(Month)[T.12]               -0.1421      0.083     -1.709      0.087      -0.305       0.021
Q("Visibility(mi)")           0.0073      0.009      0.810      0.418      -0.010       0.025
Rain                          0.3633      0.081      4.506      0.000       0.205       0.521
Q("Temperature(F)")          -0.0031      0.002     -1.824      0.068      -0.006       0.000
Q("Wind_Speed(mph)")          0.0253      0.004      6.877      0.000       0.018       0.032
Night                         0.0620      0.044      1.395      0.163      -0.025       0.149
RushHour                      0.0417      0.038      1.096      0.273      -0.033       0.116
Weekend                       0.1240      0.051      2.453      0.014       0.025       0.223
==============================================================================================
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Table 4 - County-Clustered Standard Errors

Logit Regression Results

Dep. Variable:	HighSeverity	No. Observations:	20000
Model:	Logit	Df Residuals:	19978
Method:	MLE	Df Model:	21
Date:	Fri, 17 Apr 2026	Pseudo R-squ.:	0.04213
Time:	19:59:39	Log-Likelihood:	-10138.
converged:	True	LL-Null:	-10584.
Covariance Type:	cluster	LLR p-value:	3.029e-175

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-1.4361	0.355	-4.046	0.000	-2.132	-0.740
Junction[T.True]	0.7113	0.249	2.860	0.004	0.224	1.199
Traffic_Signal[T.True]	-0.4921	0.054	-9.096	0.000	-0.598	-0.386
Crossing[T.True]	-1.1403	0.224	-5.083	0.000	-1.580	-0.701
C(Month)[T.2]	0.0703	0.053	1.317	0.188	-0.034	0.175
C(Month)[T.3]	0.2721	0.097	2.793	0.005	0.081	0.463
C(Month)[T.4]	0.2799	0.114	2.445	0.014	0.056	0.504
C(Month)[T.5]	0.4279	0.141	3.041	0.002	0.152	0.704
C(Month)[T.6]	0.4243	0.115	3.685	0.000	0.199	0.650
C(Month)[T.7]	0.3284	0.111	2.953	0.003	0.110	0.546
C(Month)[T.8]	0.3519	0.172	2.044	0.041	0.014	0.689
C(Month)[T.9]	0.2622	0.101	2.585	0.010	0.063	0.461
C(Month)[T.10]	0.2039	0.126	1.618	0.106	-0.043	0.451
C(Month)[T.11]	-0.0916	0.093	-0.983	0.326	-0.274	0.091
C(Month)[T.12]	-0.1421	0.102	-1.398	0.162	-0.341	0.057
Q("Visibility(mi)")	0.0073	0.006	1.239	0.216	-0.004	0.019
Rain	0.3633	0.155	2.338	0.019	0.059	0.668
Q("Temperature(F)")	-0.0031	0.003	-0.900	0.368	-0.010	0.004
Q("Wind_Speed(mph)")	0.0253	0.010	2.548	0.011	0.006	0.045
Night	0.0620	0.067	0.929	0.353	-0.069	0.193
RushHour	0.0417	0.054	0.777	0.437	-0.064	0.147
Weekend	0.1240	0.046	2.674	0.007	0.033	0.215

Figure 1 - Marginal Effects of Key Variables

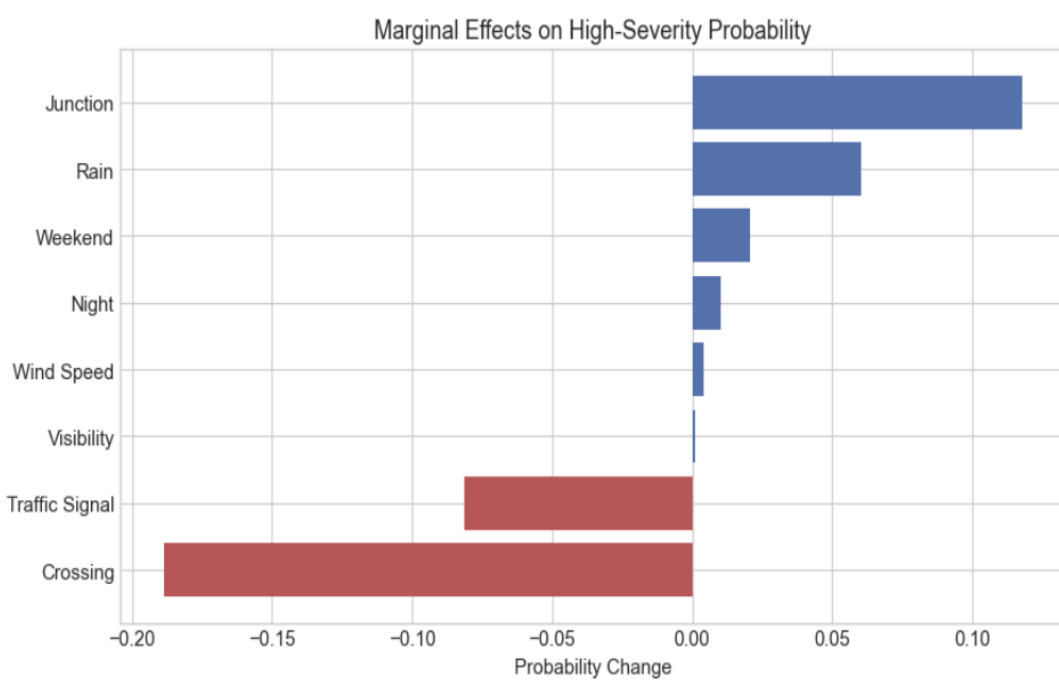


Figure 2 - High Severity Crash Rate: Rain vs No Rain

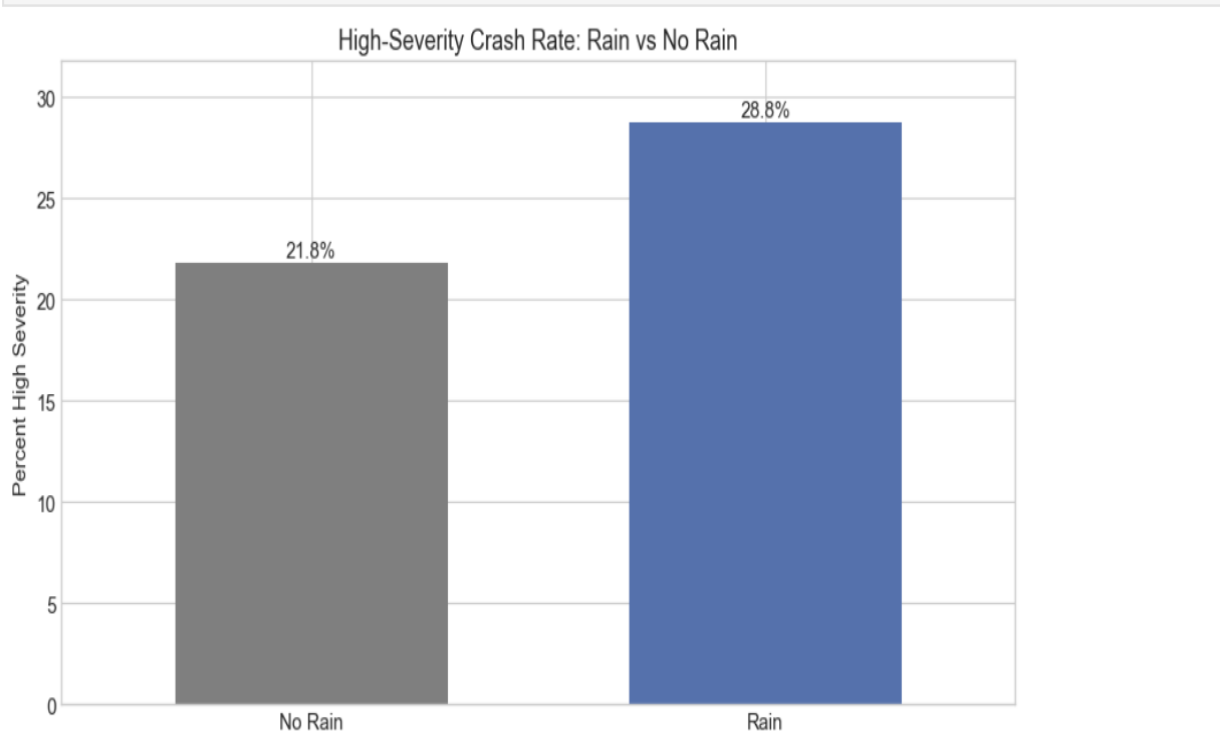


Figure 3 - Texas vs U.S. Comparison

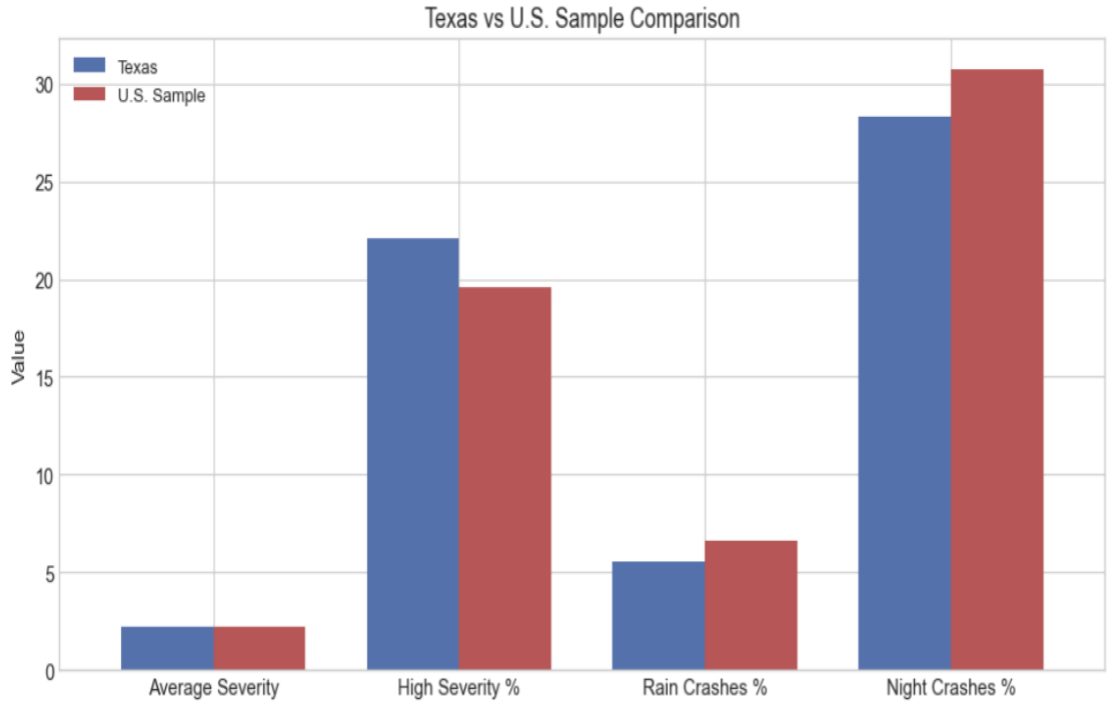
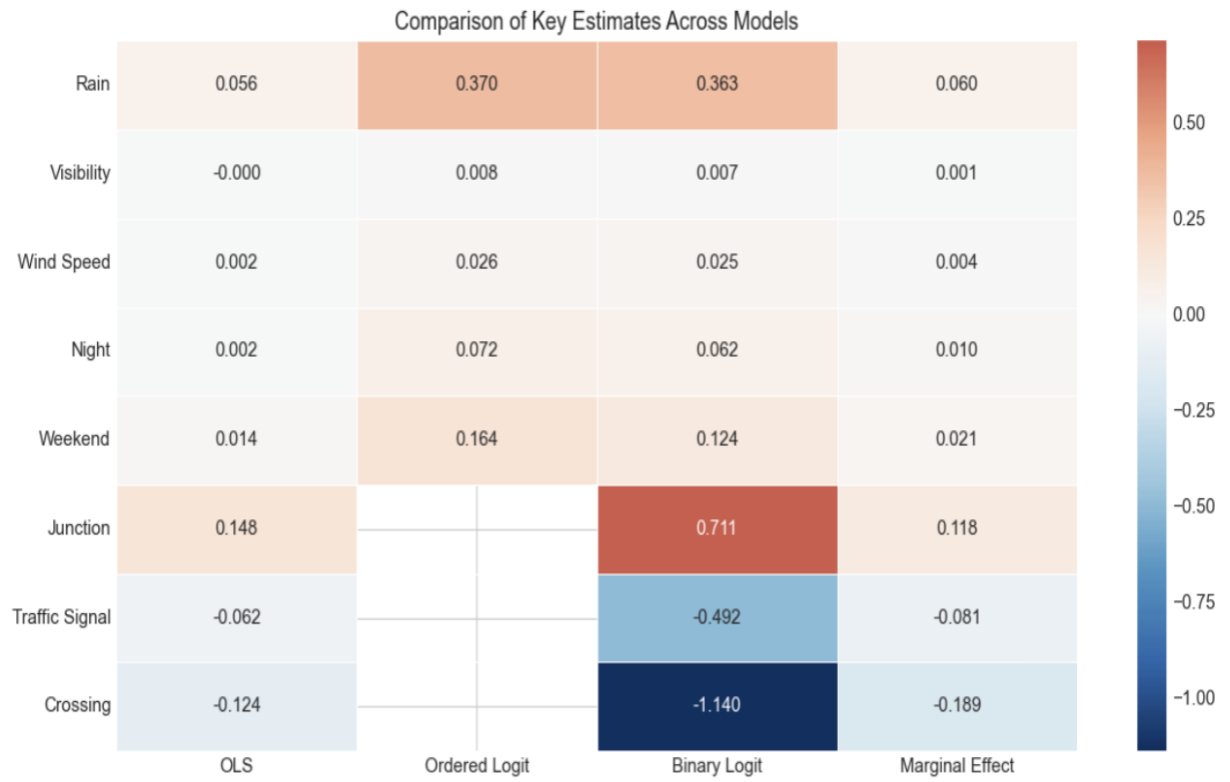


Figure 4 - Coefficient Comparison Across Models



References

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